

Research Notes on Deep Q-network

Deep Q-network

>> Section 1

3D volumes of neurons. The layers of a CNN have neurons arranged in 3 dimensions: width, height and depth. Each neuron inside a convolutional layer is connected to only a small region of the layer before it, called a receptive field. Distinct types of layers, both locally and completely connected, are stacked to form a CNN architecture. Local connectivity: following the concept of receptive fields, CNNs exploit spatial locality by enforcing a local connectivity pattern between neurons of adjacent layers. The architecture thus ensures that the learned "filters" produce the strongest response to a spatially local input pattern. Stacking many such layers leads to nonlinear filters that become increasingly global (i.e. responsive to a larger region of pixel space) so that the network first creates representations of small parts of the input, then from them assembles representations of larger areas. Shared weights: In CNNs, each filter is replicated across the entire visual field. These replicated units share the same parameterization (weight vector and bias) and form a feature map. This means that all the neurons in a given convolutional layer respond to the same feature within their specific response field. Replicating units in this way allows for the resulting activation map to be equivariant under shifts of the locations of input features in the visual field, i.e. they grant translational equivariance—given that the layer has a stride of one. Pooling: In a CNN's pooling layers, feature maps are divided into rectangular sub-regions, and the features in each rectangle are independently down-sampled to a single value, commonly by taking their average or maximum value. In addition to reducing the sizes of feature maps, the pooling operation grants a degree of local translational invariance to the features contained therein, allowing the CNN to be more robust to variations in their positions. Together, these properties allow CNNs to achieve better generalization on vision problems. Weight sharing dramatically reduces the number of free parameters learned, thus lowering the memory requirements for running the network and allowing the training of larger, more powerful networks.

>> Section 2

Convolutional deep belief networks (CDBN) have structure very similar to convolutional neural networks and are trained similarly to deep belief networks. Therefore, they exploit the 2D structure of images, like CNNs do, and make use of pre-training like deep belief networks. They provide a generic structure that can be used in many image and signal processing tasks. Benchmark results on standard image datasets like CIFAR have been obtained using CDBNs.

>> Section 3

Time delay neural networks The time delay neural network (TDNN) was introduced in 1987 by Alex Waibel et al. for phoneme recognition and was an early convolutional network exhibiting shift-invariance. A TDNN is a 1-D convolutional neural net where the convolution is performed along the time axis of the data. It is the first CNN utilizing weight sharing in combination with a training by gradient descent, using backpropagation. Thus, while also using a pyramidal structure as in the neocognitron, it performed a global optimization of the weights instead of a local one. TDNNs are convolutional networks that share weights along the temporal dimension. They allow speech signals to be processed time-invariantly. In 1990 Hampshire and Waibel introduced a variant that performs a two-dimensional convolution. Since these TDNNs operated on spectrograms, the resulting phoneme recognition system was invariant to both time and frequency shifts, as with images processed by a neocognitron. TDNNs improved the performance of far-distance speech recognition.

>> Section 4

Natural language processing CNNs have also been explored for natural language processing. CNN models are effective for various NLP problems and achieved excellent results in semantic parsing, search query retrieval, sentence modeling, classification, prediction and other traditional NLP tasks. Compared to traditional language processing methods such as recurrent neural networks, CNNs can represent different contextual realities of language that do not rely on a series-sequence assumption, while RNNs are better suitable when classical time series modeling is required.

>> Section 5

Dilation Dilation involves ignoring pixels within a kernel. This reduces processing memory potentially without significant signal loss. A dilation of 2 on a 3x3 kernel expands the kernel to 5x5, while still processing 9 (evenly spaced) pixels. Specifically, the processed pixels after the dilation are the cells (1,1), (1,3), (1,5), (3,1), (3,3), (3,5), (5,1), (5,3), (5,5), where (i,j) denotes the cell of the i-th row and j-th column in the expanded 5x5 kernel. Accordingly, dilation of 4 expands the kernel to 7x7.

>> Section 6

Animal behavior detection CNNs have been applied in ecological and behavioral research to automatically detect and quantify animal behavior from visual data, enabling identification of animals, tracking of individuals, estimation of pose, and classification of specific actions such as feeding, and social interactions. Combined with

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multi-object tracking and temporal modeling, these systems can extract behavioral sequences over extended recordings, reducing reliance on manual annotation and increasing throughput for studies of individual variation, social networks, and collective dynamics.

>> Section 7

Work by Hubel and Wiesel in the 1950s and 1960s showed that cat visual cortices contain neurons that individually respond to small regions of the visual field. Provided the eyes are not moving, the region of visual space within which visual stimuli affect the firing of a single neuron is known as its receptive field. Neighboring cells have similar and overlapping receptive fields. Receptive field size and location varies systematically across the cortex to form a complete map of visual space. The cortex in each hemisphere represents the contralateral visual field. Their 1968 paper identified two basic visual cell types in the brain:

>> Section 8

A convolutional neural network consists of an input layer, hidden layers and an output layer. In a convolutional neural network, the hidden layers include one or more layers that perform convolutions. Typically this includes a layer that performs a dot product of the convolution kernel with the layer's input matrix. This product is usually the Frobenius inner product, and its activation function is commonly ReLU. As the convolution kernel slides along the input matrix for the layer, the convolution operation generates a feature map, which in turn contributes to the input of the next layer. This is followed by other layers such as pooling layers, fully connected layers, and normalization layers. Here it should be noted how close a convolutional neural network is to a matched filter.